

of Questions: 30

Question #: 1

A 26-year-old woman comes to the physician because of a 2 week history of severe pelvic pain and urinary discomfort. Patient history shows multiple sexual partners and past treatment for gonorrhea. Pelvic examination shows pain on [palpation of the uterus. Ultrasound imaging shows uterine linings consistent with her menstrual phase. Hysterosalpingogram shows narrowing of the uterine tube. A diagnosis of salpingitis is made. Which of the following lymph nodes will most likely be enlarged as a result?

- A. Superficial and deep inguinal lymph nodes
- B. External iliac nodes
- ✓C. Para-aortic nodes at the level of the renal vessels
- D. Node of Cloquet
- E. Internal iliac nodes accompanying the uterine artery and vein

Rationale: Ovarian lymph first drains into the para-aortic nodes at the level of the renal vessels.

Question #: 2

A 23-year-old woman comes to the emergency department because of a 4-hour history of severe colicky pain in her right lower abdomen. She reports a last menstrual period at approximately 8-weeks earlier but denies pregnancy due to being on oral contraceptives. Physical examination shows a pale woman in clear painful distress. Palpation of the right iliac region produces sharp pain without rebound tenderness. A urine dipstick test shows presence of hCG. Transabdominal ultrasound shows an empty uterine cavity with an enlarged right uterine tube and fluid accumulation in the pelvic cavity. Culdocentesis is performed to access the nature of the fluid. Which of the following routes is most likely used to perform the culdocentesis?

- A. Perineal body into the vesicouterine space
- ✓B. Posterior part of the vaginal fornix into the rectouterine pouch
- C. Anterior part of the vaginal fornix into the endocervical canal

- D. Vaginal orifice into the greater vestibular gland
- E. Perineal membrane into the external urethral sphincter

Rationale: In the women, the rectouterine pouch, also known as the pouch of Douglas, represents the lowest point of the peritoneal cavity. Thus, it is the most gravity dependent allowing fluid to accumulate. The most direct route to the rectouterine pouch is through the posterior part of the vaginal fornix.

Question #: 3

A 35-year-old pregnant woman comes to the obstetrician for a scheduled ultrasound exam. Physical examination shows lower than expected fundal height for her gestational time. Ultrasonography of the uterus shows fetal growth restriction, oligohydramnios and signs of abdominal distention of the fetus. Labor is induced and postpartum examination of the neonate shows no separation between the vaginal, rectal and colonic structures. Which of the following is the most likely diagnosis?

- A. Labioscrotal folds
- ✓B. Persistent cloaca
- C. Urogenital folds
- D. Genital tubercle
- E. Urogenital membrane

Rationale: Most anorectal anomalies result from abnormal development of the urorectal septum, ultimately resulting in nondivision of the cloaca into urogenital and anorectal parts. The common outlet of the intestinal, urinary, and reproductive tracts is specifically associated with a persistent cloaca.

Question #: 4

A 34-year-old woman in active labor is brought to the emergency department. Physical examination shows ruptured membranes and 4cm dilated cervix. During the delivery the fetus does not pass through the vaginal canal as expected. During inspection of the fetal path it is noted that the umbilical cord is wrapped around the fetal arm. Assisted delivery with episiotomy is done. Prior to the episiotomy being performed the physician performs a nerve block on the area. Which of the following best describes the block procedure?

- A. Injection near ischial tuberosity anesthetizes the mons pubis and anterior labia majora
- B. Injection near ischial tuberosity anesthetizes the mons pubis and anterior labia majora
- C. Injection near ischial spines prevents pain from the uterus, cervix and upper vagina
- ✓D. Injection near ischial spines anesthetizes the posterior labia majora and inferior vagina
- E. Injection superomedial to the anterior superior iliac spine anesthetizes most perineal skin

Rationale: =The pudendal nerve is a somatic nerve that innervates the perineum with exception of the anterior part of the labium majus (innervated by the ilioinguinal and genitofemoral nerves). The pudendal nerve forms from S2 to S4 ventral rami. It leaves the pelvis via the greater sciatic foramen and enters into the perineum via the lesser sciatic foramen by coursing around the sacrospinous ligament where it attaches to the ischial spine. This pudendal nerve block will anesthetize the area of distribution of the pudendal nerve. It will not anesthetize the uterus or cervix as these are visceral organs

Question #: 5

A 62-year-old nulliparous woman comes to the physician because of an unexpected weight loss and 2 week history of abdominal and pelvic pain. Physical examination shows tenderness on palpation of the left iliac region with a hard palpable mass. Biopsy of the mass shows dysplastic tissue consistent with epithelioid ovarian cancer. Surgical removal of the mass is planned with sentinel node removal. Which of the following lymph nodes will most likely be removed during the surgery?

- A. Deep inguinal
- B. Superficial inguinal
- ✓C. Lumbar
- D. External iliac
- E. Internal iliac

Rationale: The ovaries developed anterior to the posterior abdominal wall and descended into the pelvis. As it descended, it carried its blood supply, venous and lymphatic drainage with it. As a result, the ovaries initially drain to the lumbar nodes.

Question #: 6

A 16-year-old boy is brought to the physician by his parents because he does not appear to have started puberty. Physical examination shows a boy with less than expected musculoskeletal development. Pelvic examination shows scarce pubic hair, small penis and an empty scrotal sac. Laboratory tests show lower than

expected testosterone levels. Which of the following diagnoses is most likely based on the patient's condition?

- A. Congenital deficiency of 5-alpha reductase
- B. Abdominal undescended testes resulting in cellular degradation
- C. Absence of the receptors necessary for testosterone action
- ✓D. Failure of the gonads to develop
- E. Regression of the mesonephric duct during fetal development

Rationale: DHT is converted from testosterone by 5-alpha-reductase. It is responsible for the masculinization of male external genitalia. Testosterone is produced by Leydig cells.

Question #: 7

A 21-year-old woman comes to the gynecologist for her first appointment because she has recently become sexually active. Physical examination shows normal development. Pelvic examination is unremarkable. A transabdominal ultrasound is done showing bilateral ovaries but absence of the uterine tube on the left and a smaller than expected uterine cavity. Which of the following structures most likely failed to develop resulting in her condition?

- ✓A. paramesonephric duct
- B. mesonephric duct
- C. mesonephros
- D. metanephros
- E. gonadal ridge

Rationale: Unicornuate uterus is due to incomplete development of one of the paramesonephric ducts

Question #: 8

A 3-year-old boy is brought to the physician by his mother because of a 1-month history of poor feeding and disinterest in playing. He is the fourth child in a family of six living in a war-torn region. Dietary history reveals that he is primarily fed a corn-based diet and often eats leftovers from meals with his older siblings. Physical examination shows an irritable patient with light-colored, friable hair, dermatitis, and mild bilateral pitting ankle edema. His weight is 70% of the

expected weight for age. Which of the following is the most likely diagnosis?

- A. Cachexia
- B. Marasmus
- ✓C. Kwashiorkor
- D. Stunting
- E. Anorexia nervosa
- F. Rickets

Rationale: This child presents with signs of protein-energy malnutrition, particularly kwashiorkor, which is caused by severe protein deficiency despite adequate caloric intake.

Key clinical features of kwashiorkor:

Edema: Usually pitting in the ankles due to hypoalbuminemia

Dermatitis and hair changes: Light-colored, brittle, or “flag sign” hair

Irritability and lethargy

Growth retardation: Weight often <60–80% of expected; height may be relatively preserved

Dietary history: A Diet high in carbohydrates but low in protein (e.g., corn, cassava, rice). The corn-based diet is deficient in one or more essential amino acids.

Contrast with marasmus:

Marasmus is a total calorie deficiency → severe wasting, muscle loss, no edema, skin and hair changes are less prominent

Weight is usually <60% of expected, and the child appears emaciated

In marasmus, there is severe loss of adipose tissue, thinning of arms and legs, and loss of the buccal pad of fat.

Cachexia is malnutrition due to an underlying infection or cancer. The presence of inflammatory mediators is responsible for the rapid weight loss.

Stunting leads to a reduced height. The weight for that height is appropriate.

Anorexia nervosa is usually observed in teens and is characterized by an abnormal fear of gaining weight.

Rickets is due to vitamin D deficiency before the fusion of the epiphyseal plate of the long bones. Characterized by bow legs, knock knees, thickening of ankles, and wrists. Serum ALP levels are highly elevated, with low serum calcium and phosphorus concentrations.

A 6-month-old infant girl is brought to the physician for a routine well-child visit. She has been exclusively breastfed since birth. Since her previous visit, she has gained 2.7 kg (6 lb). She is at the 80th percentile for weight and height. Which of the following best describes the nitrogen metabolism in this patient?

- A. She is in a state of nitrogen equilibrium
- ✓B. Nitrogen intake is more than nitrogen output
- C. Nitrogen intake is less than nitrogen output
- D. She is in negative nitrogen balance
- E. There is excessive muscle proteolysis

Rationale: Normal infancy is a period of positive nitrogen balance, growth, and positive energy balance.

The dietary nitrogen intake is more than the output of nitrogen in urine. There is increased muscle protein synthesis and decreased rates of muscle protein degradation.

The essential amino acids must be provided in the diet to facilitate the positive nitrogen balance and tissue protein synthesis.

Human breast milk provides an optimal balance of macronutrients for infant growth, including low but adequate protein.

Key points about nitrogen metabolism in exclusively breastfed infants:

Protein content in breast milk: ~1.0–1.2 g/100 mL

Protein is highly bioavailable and efficiently utilized for growth. Breastmilk has a higher whey-to-casein ratio.

Nitrogen from dietary protein is used for the synthesis of body proteins, with minimal waste.

Exclusively breastfed infants typically maintain positive nitrogen balance, supporting rapid growth and development.

Overfeeding protein is not necessary and may increase renal solute load; breast milk naturally prevents this.

Infants with normal growth on exclusive breastfeeding are receiving adequate protein.

Deficiency of a dietary essential amino acid can predispose to Kwashiorkor, as seen in infants with a low-quality protein diet (lacking one or more essential amino acids).

Adults are in a state of nitrogen equilibrium (nitrogen intake = Nitrogen loss in urine).

Pregnant women, adolescents, and infants are in a positive nitrogen balance.

Illnesses, such as Kwashiorkor, marasmus, and cachexia, are states of negative nitrogen balance (Nitrogen intake is less than nitrogen loss), and these patients have excessive tissue protein catabolism.

Question #: 10

A research group develops a formula for newborn infants whose mothers are unable to breastfeed. The formula is designed to mimic the composition of human breast milk and provide essential nutrients for optimal growth and development. Which of the following nutrients is naturally present in low concentration in human milk and should not be excessively increased in formula?

- A. Lactose
- ✓B. Iron
- C. Docosahexaenoic acid (DHA)
- D. Leucine
- E. Human milk oligosaccharides (HMO)

Rationale: Iron is found in low concentrations in breast milk. Iron supplementation in breastfed infants usually begins at 6 months of age to prevent iron deficiency anemia.

All other components are in human milk and must be added to the formula.

Lactose content is high in breast milk and provides glucose and galactose, which supply energy for the infant.

DHA is an omega-3 fatty acid that helps develop the brain and vision.

HMOs facilitate the growth of probiotic bacteria in the newborn gut.

Leucine is an essential amino acid and facilitates tissue growth (Positive nitrogen balance) in infancy.

Protein content in breast milk is also lower than that of cow's milk's.

Question #: 11

A 4-year-old boy is brought to the physician by his parents for a routine well-child visit. His height and weight are below the average for children of his age. He performs well in school and has normal cognitive function. The parents report that he has been a "picky eater" since age 2, consistently avoiding eggs, meat, and beans. Physical examination shows normal skin and hair, no edema, and he appears active and playful. Which of the following best describes the underlying nutritional condition in this patient?

- A. Childhood obesity
- ✓B. Stunting
- C. Cachexia

D. Kwashiorkor

E. Marasmus

Rationale: This is a child with stunting of growth. Children with Kwashiorkor have pedal edema, skin lesions and discolored hair and protuberant abdomen. Children with marasmus have a severe weight deficiency, thin arms and legs, severe loss of muscle and adipose tissue in cheeks and buttocks. Obese children have a higher BMI than normal

Question #: 12

A 6-month-old girl is brought to the physician for a routine well-child visit and immunizations. Her weight and length are appropriate for her age, and her developmental milestones are consistent with her age. The mother reports exclusive breastfeeding since birth and plans to continue breastfeeding until the child is 1 year old. She is seeking advice to begin introducing semi-solid foods. Which of the following nutritional recommendations is most appropriate for this infant?

- A. Continue exclusive breastfeeding until 1 year of age
- B. Discontinue breastfeeding and start formula feeding
- ✓C. Begin iron-containing complementary foods while continuing breastfeeding
- D. Start complementary feeding with iodine supplements
- E. Start complementary feeding with calcium supplements

Rationale: Iron-rich food must be included in an infant's diet at 6 months; otherwise, the infant is at risk of developing iron deficiency anemia. The fetus stores iron in the 3rd trimester of pregnancy. These iron stores are generally sufficient for the first six months. If the mother has an iron deficiency during pregnancy, the infant may need earlier iron supplementation.

Recommendations:

Continue breastfeeding until at least 12 months of age.

Introduce iron-rich foods such as iron-fortified cereals, pureed meats, and vegetables around 6 months.

Vitamin D supplementation should continue in exclusively breastfed infants.

Calcium and iodine supplements are not routinely advised.

Question #: 13

A 30-year-old woman in the first trimester of pregnancy comes to the physician for

a routine prenatal visit. Her BMI is 22 kg/m². She has no personal or family history of major medical illnesses and takes no medications. Physical examination is unremarkable. Which of the following nutritional recommendations is most appropriate for this patient at this stage of pregnancy?

- A. An additional 2000 Cals are required from the second semester onwards
- B. Iron supplementation prevents the occurrence of neural tube defects in the fetus
- C. Alcohol consumption during pregnancy is associated with fetal macrosomia
- D. Maternal iodine deficiency increases the risk of anemia in the newborn
- ✓E. Folic acid requirements increase during pregnancy

Rationale: Folic acid requirement doubles during pregnancy; Prenatal and perinatal maternal folate deficiency increases the risk of fetal neural tube defects. Elevated maternal serum alpha-fetoprotein (AFP) in maternal serum screening indicates fetal neural tube defects.

An additional 70, 260, 500 calories are required in the first, second, and third trimesters. Iron supplementation is crucial in the third trimester. Iron supplementation prevents iron deficiency anemia in the mother, fetus in utero, and infant after birth.

Alcohol use during pregnancy increases the risk of fetal alcohol syndrome.

Gestational diabetes increases the risk of fetal macrosomia.

Maternal iodine deficiency increases the risk of spontaneous abortion, perinatal mortality, congenital disabilities, neurological disorders, and congenital hypothyroidism.

Question #: 14

A 1-month-old infant assigned male at birth is brought to the physician for a follow-up examination. Physical examination shows ambiguous genitalia with micropenis and incomplete virilization of the external genitalia. The testes are descended bilaterally. Targeted genetic testing identifies a pathogenic mutation in the gene encoding 5-alpha reductase. Which of the following processes is most likely defective in this patient?

- ✓A. Conversion of testosterone to dihydrotestosterone
- B. Conversion of testosterone to estrogen
- C. Formation of T3 from T4
- D. Formation of 1, 25 dihydroxycholecalciferol from vitamin D
- E. Conversion of pregnenolone to dehydroepiandrosterone
- F. Formation of cortisol

G. Formation of aldosterone

H. Formation of epinephrine from tyrosine

Rationale: 5-alpha reductase converts testosterone to dihydrotestosterone.

Review the enzymes required for the listed biosynthetic reactions.

Aromatase converts testosterone to estrogen.

T3 and T4 synthesis requires thyroperoxidase and 5'-deiodinase;

Formation of 1,25-DHCC from vitamin D requires 25-hydroxylase (liver) and 1-hydroxylase (Kidney).

Epinephrine synthesis from tyrosine requires tyrosine hydroxylase (BH₄), DOPA decarboxylase (PLP), Dopamine hydroxylase (Dopamine oxidase) (Vitamin C), and phenylethanolamine N-methyl transferase (PNMT; SAM).

DHEAS formation from pregnenolone requires CYP 17 (17 hydroxylase).

Cortisol synthesis from pregnenolone requires 3-beta-hydroxy steroid dehydrogenase, CYP 17, CYP 21A, and CYP 11B1.

Aldosterone synthesis from Pregnenolone requires 3-beta-hydroxy steroid dehydrogenase, CYP 21A, and CYP 11B1 and CYP11B2 (Aldosterone synthase).

Question #: 15

A 29-year-old woman, G1P0, comes to the physician at 11 weeks' gestation for routine prenatal testing. She has no significant medical history and takes only prenatal vitamins. A noninvasive prenatal screening test shows an increased proportion of DNA fragments derived from chromosome 21, raising concern for fetal trisomy 21. The patient is counseled about confirmatory diagnostic options. Which of the following maternal samples was used for this screening test?

- A. Amniotic fluid obtained by amniocentesis
- B. Chorionic villi obtained by transcervical biopsy
- C. Percutaneous umbilical cord
- D. Paternal urine sample
- ✓E. Maternal blood draw
- F. Paternal blood draw
- G. Maternal urine sample
- H. Cervical mucus

Rationale: During fetal development, cells from the fetus (placental trophoblasts) are continuously shed and delivered to the maternal circulation where the cells are degraded and the biomolecules are recycled, or disposed of. Small snippets of fetal DNA stays in circulation for a brief amount of time (half-life of less than an hour; some estimates at about 15 minutes). Therefore, a standard maternal blood draw from a pregnant woman contains both her DNA, and

her baby's The fetal DNA can be sequenced (by next-generation sequencing technology). A typical fetus which has 22 pairs of autosomes will have equal representation of sequence reads from the cell-free DNA from all of these chromosomes. A fetus who has Down syndrome will have an extra copy of chromosome 21; and the ratio of chromosome 21 to the rest of the chromosomes will be 3 to 2. This change in the ratio of sequence reads for chromosome 21 indicates that the fetus likely has Down syndrome.

Read more from the website <https://news.sanfordhealth.org/womens/cell-free-dna-testing/> ... Cell-free DNA is a new lab test offered during pregnancy that is used to screen for Down syndrome. This test is also called NIPT, which stands for non-invasive prenatal test because it involves simply taking a sample of blood from the mom's vein rather than inserting a needle into the uterus like an amniocentesis. Test results are generally available within seven to 10 days. NIPT has great potential as a screening test for genetic disorders in pregnancy, but there are some limitations that must be understood by patients so the test is used properly.

Question #: 16

A 6-month-old infant is brought to the physician by his parents because of concerns about abnormal limb development. Physical examination shows shortened proximal limbs, an enlarged head with frontal bossing, and midface hypoplasia. Radiographs reveal abnormal organization of cartilage growth plates with disordered chondrocyte architecture. Genetic testing identifies a mutation in the FGFR3 gene. Which of the following terms best describes the underlying abnormality in this patient?

- A. Sequence
- B. Association
- ✓C. Dysplasia
- D. Deformation
- E. Disruption

Rationale: A dysplasia is defined as the presence of cells of an abnormal type within a tissue; in patients with developmental disorders, this occurs during embryogenesis and fetal development. A sequence describes how one primary defect can cause secondary problems; a classic example is the Robin sequence where a small lower jaw (micrognathia) causes abnormal placement of the tongue, which leads to cleft palate. An association describes how a constellation of findings might be found in a patient more often than chance alone would predict, but no known single genetic cause is known. A deformation occurs when extrinsic mechanical force impacts on a normally developing or developed structure to cause a dysmorphology. An example would be non-syndromic clubfoot (talipes) or congenital hip dislocation – note that in the case of the deformation, a genetic factor is often not involved. There is a subtle difference

between a disruption and a deformation. A disruption is a morphologic malformation of a part of the body, an organ, or a part of an organ, caused by interference with an originally normal developmental process. An example would be amniotic band disruptions that can cause a digital amputation.

Question #: 17

A 28-year-old woman, G2P0, goes to her physician for prenatal counseling. Her previous pregnancy resulted in an infant with severe holoprosencephaly who died shortly after birth. She denies teratogenic exposures during that pregnancy. Physical examination reveals a single maxillary central incisor. Fetal ultrasound in the current pregnancy is normal so far. Which of the following genes is most likely affected by a loss-of-function mutation in this patient?

- ✓A. *SHH*
- B. *FGFR3*
- C. *CFTR*
- D. *SHOX*
- E. *MYC*

Rationale: The Sonic Hedgehog (*SHH*) gene encodes a soluble morphogen protein which initiates signaling through a pathway that regulates cell fate to control midline development (craniofacial) and also limb bud formation. Haploinsufficiency of *SHH* can cause: no defect; a single tooth incisor; cleft lip/palate; or holoprosencephaly. This is an extreme example of variable expressivity, which is well documented for this gene. Interestingly, this variable expressivity has even been documented among members of the same family. *FGFR3* is a fibroblast growth factor receptor (tyrosine kinase). Mutations in *FGFR3* that cause unregulated signaling of this receptor cause premature ossification of the growth plates of the long bones during development leading to the classic dwarfism phenotype of achondroplasia. *CFTR* encodes a chloride channel; autosomal recessive inheritance of *CFTR* mutations causes CF. *SHOX* is a transcription factor that is important to control attainment of normal stature. *SHOX* is an example of a gene which is found on both the X, and Y chromosomes (it is found on the pseudoautosomal region of the Y). Since *SHOX* is also on the Y chromosome, it therefore must escape X-inactivation. Haploinsufficiency of *SHOX* is thought to cause the short stature seen in Turner syndrome. *MYC* encodes a transcription factor that regulates the expression of many genes that are required for transit through the cell cycle. Because of this central role in preparing for cell cycle progression, overexpression of *MYC* is a hallmark of the cancer genome.

Question #: 18

A 25-year-old woman delivers a healthy-appearing newborn boy at term without complications. On day 2 of life, routine newborn screening reveals markedly elevated blood phenylalanine levels. The infant is started on dietary therapy while confirmatory testing is arranged. The mother's pregnancy and delivery were unremarkable, and there is no family history of metabolic disease. Which of the following diagnostic tools was most likely used to detect this abnormality?

- ✓A. Mass spectroscopy
- B. cDNA microarray
- C. Karyotype
- D. Decreased AFP
- E. Southern blot

Rationale: Mass spectrometry (MS) can be used to identify thousands of metabolites in a biological sample. In the most simplistic terms, the MS instrument delivers some energy to a molecule which causes it to break and attain a charge (ionization). The molecular fragment is then accelerated. Newtonian physics tells us that the velocity of a molecule is dependent upon its mass. Back calculating from the charge and the mass of the molecular fragments allows identification of the molecule of interest. Alpha Feto Protein (AFP) levels are useful in the prediction of a few specific fetal anomalies such as spina bifida, anencephaly, and some aneuploid conditions. The other tests mentioned are used for the analysis of DNA, not metabolites.

Question #: 19

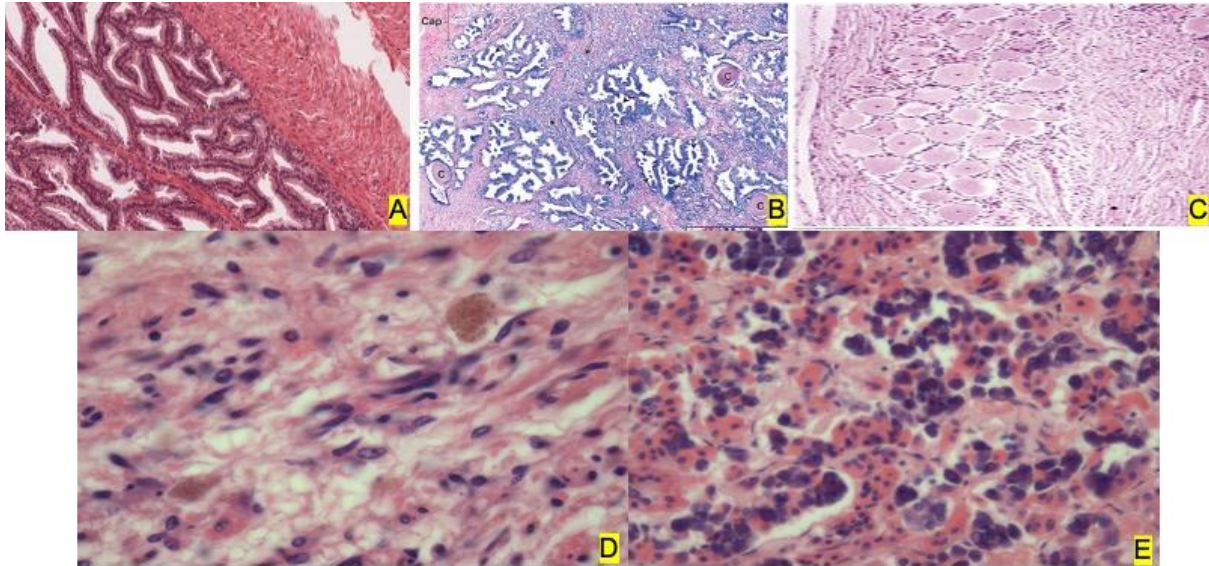
A 32-year-old woman is brought to the emergency department because of a 1-hour history of vaginal bleeding. She is at 40 weeks' gestation. Physical examination shows the absence of uterine contractions. The physician suspects a deficiency of the hormone responsible for the promotion of uterine contraction during delivery. Which of the following photomicrographs best corresponds to a structure most likely affected to result in this hormonal deficiency?

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: A - Seminal vesicle
B- Prostate

- C - Sensory ganglion
- D- Posterior pituitary (correct answer)
- E - Anterior pituitary

The hormone oxytocin is responsible for uterine contractions. Oxytocin is produced by the paraventricular nuclei of the hypothalamus, and travel down the hypophyseal tract to get stored in the posterior pituitary. If there are injuries anywhere in this trajectory, then the production and release of this hormone will be impacted, such as with this case. Of all the images presented, injury to the posterior pituitary is most consistent with a deficiency of this hormone.



Question #: 20

A 39-year-old man comes to the physician because of a 1-year history of inability to conceive. His vital signs are within normal limits. Physical examination shows no abnormalities. A biopsy of his testes show distorted morphology of a columnar shaped cell with a pale nucleus and prominent nucleolus. Which of the following is most likely a laboratory finding in this patient's condition?

- A. Low testosterone
- B. Low LH
- C. Elevated inhibin
- ✓D. High FSH levels
- E. Elevated ABP levels

Rationale: The described cell is a Sertoli cell. These cells have receptors for FSH, and also produce inhibin as well as ABP. If these cells are damaged, then there would naturally be

lower levels of ABP and inhibin and FSH levels would be elevated as the feedback mechanism would be affected.

LH is produced by the gonadotrophs of the anterior pituitary gland, and acts on the Leydig cells to produce testosterone. However, these cells are not affected, and therefore testosterone levels would be normal. LH levels would also be normal because the anterior pituitary gland is not affected.

Question #: 21

A 24-year-old woman is brought to the emergency department by her husband because of a 1-week history of intermittent vaginal bleeding. She is at 12 weeks' gestation. Physical examination shows abdominal tenderness. Ultrasound shows an intrauterine mass. Histopathological examination of a biopsy specimen shows proliferation of cytotrophoblasts and syncytiotrophoblasts deep into the endometrium. Which of the following hormones is most likely being produced by these trophoblastic cells?

- A. Prolactin
- B. Oxytocin
- ✓C. hCG
- D. Melatonin
- E. Cortisol

Rationale: Chorionic villi are composed of a central core of mesenchyme surrounded by an inner layer of cytotrophoblasts and an outer layer of syncytiotrophoblasts. These cells produce an array of hormones including hCG which helps to maintain the pregnancy.

Prolactin is produced by the anterior pituitary (lacto/mammotrophs).

Oxytocin is produced by nuclei of the hypothalamus and stored in the posterior pituitary

Melatonin is produced by cells of the pineal gland.

Cortisol is produced by the spongiocytes of the zona fasciculata in the cortex of the adrenal gland

Question #: 22

A 32-year-old woman comes to the physician because of a 2-day history of a lump in her right breast. She is 6 months postpartum, and has stopped breast feeding 2 months prior. Physical examination shows a well-defined, hypermobile 2.5 cm x 2.5 cm mass. A biopsy specimen is shown in the attached photomicrograph.

Which of the following will most likely be a normal finding on biopsy in this patient?

- A. Lots of stromal tissue
- ✓B. Well-developed glands
- C. Underdeveloped glands
- D. Glandular atrophy
- E. Very little amount of adipocytes

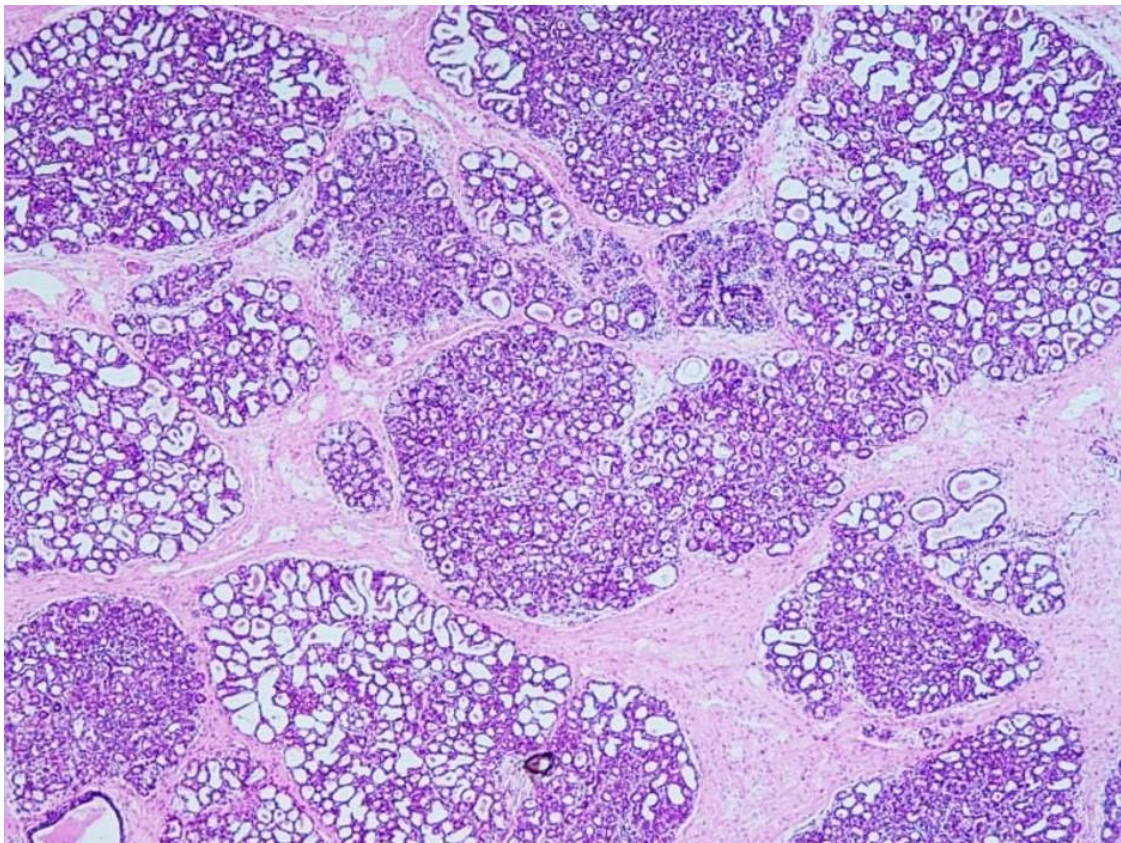
Rationale: The image shows clusters of glandular tissues (lobules) with well developed acini filled with secretion and surrounded by a small amount of connective tissue. These features are characteristic of the terminal duct lobular units (TDLU's) in actively secreting mammary glands. This woman has stopped breast feeding 2 months ago, so she is more than likely in the proliferative stage of an active mammary gland as opposed to the lactating phase. The major difference would be scant amounts of connective tissue in a lactating breast. During pregnancy and lactation TDLU's proliferate, acini appear and secretions of milk components may be found in the lumen.

A - present in pre-pubertal and pubertal stages. In the pubertal stages, more adipocytes appear.

C - present in the pre-pubertal stages

D- present in menopausal states

E - present in the pre-pubertal states



Question #: 23

A 28-Year-old-man comes to the physician because of a 1-year history of infertility. He was diagnosed with lymphoma and received chemotherapy 18 months ago. His vital signs are within normal limits. Physical examination shows normal male secondary sex characters, normal testes and penis. Laboratory studies show low sperm count but normal serum testosterone level. He is suspected of having Sertoli cell dysfunction. Which of the following serum hormonal alterations are most likely to be seen in FSH and LH levels respectively?

- A. Decrease, decrease
- B. Normal, normal
- C. Normal, increase
- ✓D. Increase, normal
- E. Increase, increase

Rationale: Sertoli cells produce Inhibin Which suppresses FSH production from the anterior Pituitary. There would be no Inhibin production with complete selective destruction of the Sertoli cells, leading to increased FSH levels. The Leydig cells are unaffected, and the remainder of the hypothalamic Pituitary gonadal axis is intact; therefore, we would not expect changes in testosterone or LH levels.

Question #: 24

A 31-year-old woman at 28 weeks' gestation comes to the physician for a routine examination. This is her first antenatal visit. Her pulse is 82/min, blood pressure is 128/84 mm Hg, and respirations are 16/min. Physical examination shows normal fetal heart sounds. Laboratory studies show elevated plasma estrogen, progesterone, and prolactin levels. Which of the following is most likely decreased during this stage of gestation?

- ✓A. Hematocrit

- B. Blood volume
- C. Fibrinogen
- D. Cardiac output
- E. Tidal volume

Rationale: During pregnancy, there is an increase in fibrinogen, blood volume, cardiac output and tidal volume. There is rapid retention of sodium and water and blood volume expansion. However, this is not accompanied by a rapid increase in the red cell mass which results in hemodilution and decrease in hematocrit.

Question #: 25

A 31-year-old woman at 24 weeks' gestation comes to the physician for a routine examination. She feels good and does not have any complaints. Her pulse is 86/min, blood pressure is 130/85 mm Hg, and respirations are 18/min. Physical examination shows normal height of the uterine fundus for the gestational age and normal fetal heart sounds. She tells the physician that she would like to know more about the placental functions. Which of the following is true regarding the function of the placenta?

- A. Allows passage of all immunoglobins
- B. It is a perfect barrier that prevents infections from transmitting to the fetus
- ✓C. It has strong aromatase activity, which converts androgens to estrogens
- D. It is the major source of progesterone during the first six weeks of pregnancy
- E. It produces human placental lactogen (HPL), which increases maternal Insulin sensitivity

Rationale: The placenta produces many hormones, including estrogen and progesterone. It has strong aromatase activity converting androgens to estrogens, preventing masculinization of the female fetus (Choice C). The Corpus luteum is the major source of progesterone during the first six weeks of pregnancy (Choice D). The placenta confers passive immunity to the fetus but only allows passage of IgG (Choice A). The placenta is not a complete barrier against infections; many viruses and bacteria can cross the placenta and enter the fetal circulation. Lastly, the placenta produces Human Placental Lactogen (HPL), which decreases maternal insulin sensitivity allowing more glucose to be available to the fetus.

Question #: 26

A 16-year-old girl is brought to the clinic by her mother for a routine examination. Her menarche occurred at the age of 11 years. Her temperature is 37°C (98.6°F), pulse is 86/min, blood pressure is 118/ 75 mm Hg, and respirations are 16/min.

Physical examination of the breast shows that the papilla and the areola project to form a secondary mound above the rest of the breast tissue. Her pubic hair resembles adult type and is distributed in the classic female triangular pattern and extends on her medial thighs. Which of the following tanner stages of puberty is this patient most likely in?

- A. Stage 5 for both breast and pubic hair development
- B. Stage 4 for both breast and pubic hair development
- ✓C. Stage 4 for breast development and stage 5 for pubic hair development
- D. Stage 4 for pubic hair development and stage 5 for breast development
- E. Stage 3 for breast development and stage 5 for pubic hair development

Rationale: She has stage IV development of breast because areolae and papillae project to form a secondary mound above the rest of the breast tissue. At stage, V development, the secondary mound disappears, only the papillae project above rest of the breast tissue. Regarding pubic hair, she has stage V (adult type), as some pubic hairs extend to medial thigh.

Question #: 27

A 47-year-old woman comes to the physician because of a 1-year history of missed periods and episodes of anxiety and irritability. Her menses previously occurred at a regular 28-day interval. She is sexually active with one male partner and uses condoms consistently. She is 163 cm (5 ft 4 in) tall and weighs 80 kg (176 lb); BMI is 30.1 kg/m² (Normal: 19-25 kg/m²). Her vital signs are within normal limits. Physical examination shows vaginal dryness and atrophy of the vaginal mucosa. Elevated serum level of which of the following hormones most likely confirms the diagnosis of menopause in this patient?

- A. Adrenal androgens
- ✓B. Follicle-stimulating hormone
- C. Human chorionic gonadotropin
- D. Adrenocorticotrophic hormone
- E. Inhibin

Rationale: A woman is formally defined to be in menopause 12 months following the completion of her final menstrual period. This patient presentation is consistent with menopause. Supporting features include her age, amenorrhea for 12 months, irritable, anxious, vaginal dryness and vaginal mucosal atrophy. In menopause the ovaries atrophy and are characterized by the presence of few to no healthy follicles. The decline in ovarian function is associated with a decrease in estrogen secretion and a concomitant increase in LH and FSH, which can be used

as a diagnostic tool.

Question #: 28

A 39-year-old woman comes to the physician because of a 3-month history of missed periods. She is wondering if she is menopausal. She also states she has breast tenderness. She is married and has 2 children. Her last child was born at age 34. Her vital signs are within normal limits and physical examination shows no abnormalities. Her urine pregnancy test is positive and her blood tests shows high levels of prolactin, estrogen and progesterone. The action of high prolactin on milk production in the breast tissue is most likely suppressed by which of the following hormones in this patient?

- A. Oxytocin
- ✓B. Estrogen and progesterone
- C. Gonadotropin releasing hormone
- D. Luteinizing hormone
- E. Follicular stimulating hormone

Rationale: High levels of estrogen & progesterone prevent the milk synthesis actions of high levels of prolactin on breast. Oxytocin is required for milk ejection. GnRh

Question #: 29

A 24-year-old woman comes to the physician because of a 2-month history of missing menstrual periods. She says that she was taking birth control pills over a year ago but stopped about 6 months ago. Her vital signs are within normal limits. Physical examination shows no abnormalities. The physician suspects that she may be pregnant and requests a pregnancy test. Which of the following laboratory findings are most likely to confirm the suspected diagnosis?

- A. Low plasma progesterone and high luteinizing hormone
- B. High prolactin, low luteinizing hormone, and low progesterone
- C. High urinary estradiol and low progesterone
- ✓D. High urinary human chorionic gonadotropin and high plasma progesterone
- E. High follicle stimulating, luteinizing hormone and human chorionic gonadotropin

Rationale: She is at 8 week pregnancy when corpus luteum of pregnancy secretes

progesterone and trophoblastic cells secrete hCG.

Question #: 30

A 25-year-old woman comes to the physician for a routine examination. She got married recently and wishes to have a child. Menarche occurred at the age of 12 years and menses occur at regular 28-day intervals. She is not on any birth control pills and is not using any other methods of contraception. She takes no medication. Physical examination shows no abnormalities. Laboratory studies show a very high serum estrogen level with low baseline serum progesterone levels. Based on these laboratory results, which of the following events is most likely to occur within the next 2 days in this woman?

- A. Cessation of menstruation
- B. Decrease in basal body temperature
- ✓C. Ovulation
- D. Thickening of cervical mucus
- E. Regression of corpus luteum

Rationale: This patient is normal, and the questions basically asks at what stage of the menstrual cycle are estrogen levels high and progesterone low. Estrogen peaks in both the follicular and luteal phases, and progesterone increases only in the luteal phase. Thus the only correct menstrual phase is the follicular phase. This increase in estrogen in the follicular phase results in a switch to positive feedback and the LH surge, and thus causing ovulation.